

The Promises and Pitfalls of Using Panel Data to Understand Individual Belief Change*

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Abstract

We investigate whether studies on political belief change can identify change trajectories at the individual level. Using simulations and case studies, we propose a grid-search framework that allows researchers to evaluate the extent to which their target estimates generalize to their study population. We use simulated datasets to estimate plausible values for how many people changed, how much they changed, and who changed, based on observed response trajectories. Our results suggest that researchers should think carefully about the conditions under which they may make claims about belief change at the individual level. We provide an R package to help researchers implement this procedure in their own work.

Studies on political change at the individual level—change in one’s beliefs and preferences—face a challenging problem: while our methods typically provide estimates for change and stability at the population level, our goal is often to understand these processes at the individual level.

This *group-to-person generalizability problem*, the claim that group-level findings may not generalize to each person (McManus, Young, and Sweetman 2023), has important implications for what we can and can’t say about political change across the life-course. While the previous research quantified change with various methods—ranging from assessing the time order of longitudinal observations (Kiley and Vaisey 2020; Vaisey and Kiley 2021) to multilevel decomposition of over-time variance (Lersch 2023)—these studies have been unable to disentangle whether they find a small amount of change in a large number of people, a large amount of change in a small number of people, or a mix of positive and negative trajectories. In other words, while social science researchers have become sophisticated in evaluating the average levels of change over time in a population, they have mostly overlooked the fact that these average changes can be produced by different processes.

This issue arises in many ways in empirical research, particularly when we want to theorize the mechanisms of political belief change. For example, the General Social Survey (GSS) shows that roughly 35% of the American adults in 2006 believed that “marijuana should be legalized.” In less than 10 years, this support had increased to nearly 60%. As seen in Figure 1, this change occurred both *cross-sectionally*, where average support increased with time, and *longitudinally*, where at least some segments of the American population changed their minds on this policy over these periods.

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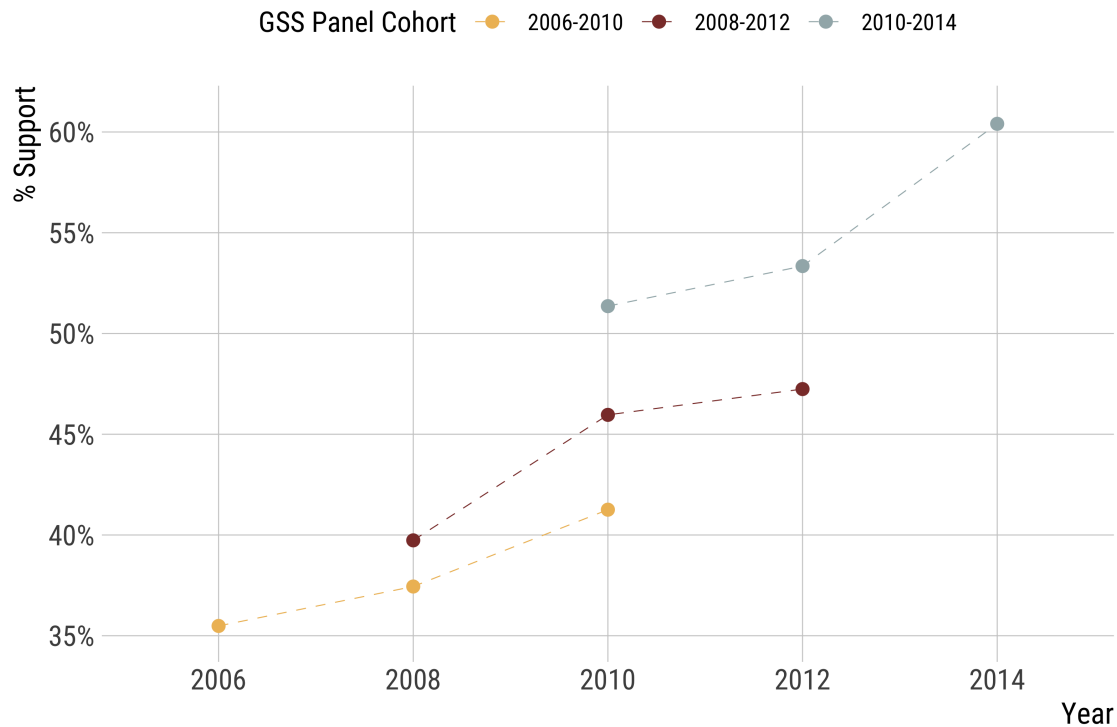


Figure 1: The Percent Support for Marijuana Legalization

Notes: The figure depicts the percent of American adults saying "marijuana should be made legal" in the General Social Survey 2006, 2008, and 2010 panel studies. We included respondents who participated in all three waves of each panel study and weighted the results using GSS post-stratification weights. Note that panel attrition caused compositional changes in the estimates, and compared to the GSS time-series data, the estimates of support are marginally inflated.

However, these two processes, cross-sectional change and longitudinal change, attract competing theories of political change, and group-level estimates may not properly adjudicate the underlying data generation processes. Following the same example, the cross-sectional change in beliefs about marijuana legalization may have occurred via two processes: people may have changed their minds following a changing *zeitgeist*, or old generations who disapproved this policy might have been replaced by new generations approving it (Ryder 1985; e.g., Small 2002). Similarly, the longitudinal change could result from various processes: large segments of the population may have shifted their opinions positively in small amounts, a small segment of the population may have made large changes, or a mix of changes—both approvals and disapprovals—may have altered the balance.

While the cross-sectional processes have received strong scholarly attention in the literature (Bartels and Jackman 2014; Ochoa and Vaisey 2024; Vaisey and Lizardo 2016), questions of the second sort remained largely unexplored. However, given that theories of political ideology involve theoretical expectations about individual-level outcomes, the fact that we lack information about whether our models capture person-level empirical processes is detrimental to understanding adult political belief change. Our article aims to address this gap by examining whether we can identify (a) *how many people changed*, (b) *how much they changed*, and (c) *who changed* using empirical data.

Using simulations and empirical case studies of political change, we show that it is indeed possible to use panel data to provide plausible estimates of which generative models are more likely to have produced particular empirical observations, even if identifying exactly which model produced the data is implausible. We also find, however, that a reliable identification of who changed occurs only under very narrow conditions. Hence, the problem of modeling individual change is very much like the age-period-cohort identification problem that arises in repeated cross-sectional data (Fosse and Winship 2019). In both cases, the problem cannot be “solved” by any particular methodological trick. Because of this, our objective is *not* to provide a methodological panacea to solve the problem, but rather to provide a concrete approach that can help us understand the issue more clearly and narrow the range of possible solutions.

This approach builds on the “Approximate Bayesian Computation” (ABC) literature (Beaumont 2010) and provides a simple grid-search procedure to help researchers evaluate the extent to which their target estimates might be generalized to a study population. Using this procedure, we show that it is possible to approximate the set of underlying data generation processes for *rate of change*, *strength of change*, and *direction of change* in a panel dataset. We provide an R package, “gridsearch,” to help researchers implement our approach in their own empirical applications.

We emphasize the existence of striking heterogeneity in these underlying generative processes even for a seemingly straightforward case like the marijuana legalization in the GSS; that is, we show that theoretically opposite claims—e.g., large swaths of the population increasing their support for the policy, as opposed to a small group of people completely switching their positions—may very well account for the same data. This finding is particularly significant in the longitudinal panel context, which, for most applications, represents the gold standard in evaluating change in observational research. For this reason, we argue that when the meaning of “political change” is not clearly specified, our current empirical strategies could potentially mislead rather than inform, especially if we overlook the underlying generative mechanisms.

We begin with a general overview of the “latent variable model” of beliefs, the problems of reliability and resolution in survey measurement, and the compositional sources of aggregate change. We then introduce our grid-search approach with an illustrating exercise. We provide empirical examples and explore the implications of our approach. Next, we analyze whether our simulation approach helps us in classification tasks, particularly in evaluating *who* changed. We conclude by discussing the promises and pitfalls of panel data for understanding individual political change.

Measuring Change in Political Beliefs and Preferences

The latent-variable approach to survey response is based on the assumption that there is a random variable Y , which captures a person i ’s “true” position in a continuous latent distribution (Alwin 2007). This random variable might stand in for any disposition, including one’s political beliefs and preferences, which allows us to define one’s political position as their *true score* Y_i .¹

The latent-variable approach acknowledges a fundamental problem in survey research: any measure of one’s true score Y_i is almost always realized with a certain degree of measurement error. Call this realization the *observed score*, y_i . We often assume that y_i represents the expected value of

¹Naturally, the assumption that *there is indeed a true score* might miss cases where individuals hold contradictory ideas (Swidler 1986) and are ambivalent about their true positions (Zaller 1992; Zaller and Feldman 1992). In this article, we follow the latent variable model and assume that there is a true position for each individual; hence, bracketing the cases where this ambivalence needs to be incorporated into the survey response measurement model.

hypothetical draws from Y_i , with an ϵ term reflecting the measurement error in this process—i.e., $y_i = Y_i + \epsilon_i$. This error might result from various factors that introduce biases in how we extract information from individuals, given that problems like question construction and interviewer effects might influence how individuals reveal their true preferences (Alwin and Krosnick 1991). The inherent randomness in the measurement process implies that each draw from Y_i might at least slightly differ from the true score, and this error term is difficult to circumvent with a one-item measure, which is often the case in large public opinion surveys. Therefore, there is often a degree of systematic gap between Y_i and y_i .

In the panel context, where we observe the same people multiple times, this measurement problem has substantive implications: if true scores are imperfectly realized and we only have access to a person’s observed survey responses, how can we know if they *actually* changed between time t and $t + 1$, or whether the observed change is merely due to measurement error? For instance, when we claim that support for abortion “rose an average of 28 points by mid-1970s” (Hout, Perrett, and Cowan 2022:2), or preferences for redistribution changed among 50% of the GSS respondents who became unemployed between 2006 and 2008 (Owens and Pedulla 2014), how can we be sure that these changes represent true changes in people’s latent positions?

Scholars in public opinion research have attempted to address this issue in multiple ways, using structural equation models that explicitly quantify the measurement error component (Achen 1975), strategies that integrate longitudinal surveys to exploit the information in the ordering of responses (Kiley and Vaisey 2020), and various multilevel mixed-effects models that shrink responses based on population estimates to derive reliable trends (Hout and Hastings 2016; Lersch 2023). To some extent, all these strategies have been useful for offsetting the problems associated with measurement error.² That said, while these empirical strategies have been largely successful in accounting for the amount of change *in the population*, they were rather silent about changes *at the level of individuals*.

The Compositional Sources of Aggregate Change

Suppose we have a panel data, where individuals $i = 1, \dots, N$ are observed over two time periods t_0 and t_1 . Let y_{it} represent an observed survey response. To see how much observed change there is at the individual level, we can simply define a change quantity, $\delta y_{i,t_0 \rightarrow t_1} = y_{it_1} - y_{it_0}$, the average of which, $\bar{\delta}_y$, represents the mean change in the dataset from t_0 to t_1 .

This quantity is a natural candidate for describing aggregate belief change, but, under the hood, it summarizes three compositional processes in one metric. Let us define these processes intuitively. The first component is R , or the *rate of change*. This represents the proportion of individuals in our dataset with $\delta y_{i,t_0 \rightarrow t_1} \neq 0$, i.e., the proportion of individuals who changed in this period. The second component is S , the *strength of change*. This represents the average magnitude of change among those who changed.³ The third component is D , the *direction of change*. This represents the net directionality of change, i.e., the difference in proportions of positive and negative changers in our observations. If we combine these three components, we can represent our aggregate change measure, $\bar{\delta}_y$, simply as

²However, note that the use of repeated observations in the panel context comes with additional assumptions: it is rather straightforward to assume that multiple measures at time t can be indicative of an underlying construct—e.g., we can average them to have an index—but the use of different time periods needs a *stability* assumption, that is, the reliability estimates may only be computed if one assumes that people do not really change between waves, which might confuse true change with measurement error (see, e.g., Hout and Hastings 2016).

³Of course, we might further decompose this average strength to a distribution of heterogeneous change trajectories across individuals. For analytical and expository clarity, we bracket this issue in this article.

$$\bar{\delta}_y = R \times S \times D$$

This simple decomposition should let us appreciate that the average change score may be generated via multiple data generation processes, each having strongly different theoretical implications.

For instance, if we want to evaluate how a society became more pro-redistribution over time, we need to note that an increase in pro-redistributive preferences could occur via all or most people reporting higher preference for redistribution, some people switching their positions altogether while some others remain completely stable, or everyone changing in one or the other direction, and the net score we observe is simply the difference in their overall trajectories. The same reasoning applies to the GSS’s marijuana legalization policy we discussed above. Most discussions in political belief change—confidence in legislative groups or financial institutions, beliefs about the economy, or gender attitudes—have the exact same question about their data generation processes. Simply put, the findings about individual change at the population level may simply result from a variety of incompatible, though observationally equivalent, data generation processes. Again, this is very similar to the age-period-cohort (APC) identification problem we mentioned above.

If we were to believe that *each and every single survey response* represents an accurate mapping of true scores and observed scores, we could seamlessly solve this problem by simply looking at the empirical data. But we contend that the measurement problem does not allow us to do that. The process often fails in at least two ways: (1) our observed measures might have low *reliability*, such that a hypothetical set of repeated draws may significantly differ from one’s true score, and (2) the question response categories may be imperfect: the *resolution* of responses may fail to distinguish consequential differences in true scores. Therefore, the average change captured in $\bar{\delta}_y$ may represent a biased or imperfect estimate of change to begin with.

We can now clearly state our research question: given a longitudinal panel study where we observe some individuals across time, how can we determine the underlying generative process—that is, how many people changed, how much their latent positions changed, and who changed—given a level of latent score reliability, response resolution, and a longitudinal survey design?

We pose this question in a fixed-effects longitudinal panel data context (Allison 2009; Vaisey and Miles 2017), with varying individual trajectories across time periods (Rüttenauer and Ludwig 2020). This allows us to shift the focus from *aggregate change* to *individual change*, given that our theoretical interest is to understand how people change their political beliefs over the life-course.

A Grid-Search Procedure to Adjudicate Data Generation Processes

We build on the Approximate Bayesian Computation (ABC) literature (Beaumont 2010; Sisson, Fan, and Beaumont 2018) and propose a grid-search procedure to answer our research questions. ABC is a simulation-based approach to parameter estimation that follows a Bayesian logic.

We implement this approach using a rejection sampler algorithm following these steps:

- 1) Create a summary function. In our case, we use the distribution of individual-level change scores to summarize the observed data. More precisely, in the case of a binary outcome, we fit

$$\mathbb{P}(y = 1) = \beta_0 + \beta_1 t, \quad t \in [0, 1]$$

where t is a normalized time variable, and estimate a predicted change score by calculating the difference between the prediction at $t = 1$ and the prediction at $t = 0$.

- 2) Simulate data with several parameters of interest for our DGPs—rate of change, strength of change, direction of change, and reliability of response measurement—using

$$y_{it} = Y_i + \delta\tau_i D_{it} + \epsilon_{it} \text{ with } \epsilon_{it} \sim \mathcal{N}(0, 1)$$

where we have a random variable Y_i for true scores, with varying realizations y_{it} , representing a person i 's beliefs at time t ; an indicator for change, D_{it} , operationalized as a non-reversible trigger if there is a respondent-level change in true scores; an effect size, δ , and a respondent level direction multiplier τ_i that indexes whether the change is positive or negative.⁴

- 3) Compute the distance between the summary of the observed data and the simulated one. We use the Kolgomorov-Smirnov statistic to measure the degree of overlap between the real and simulated distributions of change scores.
- 4) Replicate steps 2 and 3 N times, each one drawing new samples from a proposal distribution for each parameter.
- 5) Take the samples that resulted in the smallest distance as plausible DGPs for our observed data. Smaller distances indicate a closer resemblance of the simulated to the observed data, indicating that the proposed DGP could have generated our real-world data.

The grid-search procedure we propose is powerful: it allows us to make educated inferences about the extent to which our parameter estimates can generalize to our populations.⁵

Approximating Observed Data via Simulations

Let us flesh out our approach by walking through an illustrative example of how we approximate the observed empirical responses. Suppose that we *know* the underlying DGP of a dataset of interest. In this dataset, called DGP 1, 500 people are observed over a period of 12 waves. We know that 50% of these respondents make certain changes to their latent positions, and these changes are all in the positive direction with a strength of 1 SD. However, we also know that the item reliability is 80%, which means that, across time, we sometimes miss respondents' true scores. The top-left panel of Figure 2 shows the distribution of change scores from this dataset.

Assume that this dataset is our *observed* dataset. We will now see whether an alternative dataset, called DGP 2, can approximate its observed parameters. From DGP 1, we know that there are 500 individuals, observed across 12 waves. We also know the marginal distribution of our response variable. We can treat these parameter values fixed for our new simulation. However, we do not know several other parameters: we do not know the *real* rate of change, nor do we know the *real* strength, direction, or reliability. Let us assume, for this specific iteration, a rate of 50% (correct), a strength of 1 SD (correct), a reliability score of 80% (correct), and an equally split direction, half changing positively and half changing negatively (incorrect). The bottom-left panel in Figure 2 shows the distribution of change scores from this particular DGP.

⁴Note that in this model, actors can change at any wave of the study, making D_{it} variable over t .

⁵We define additional parameters that we fix in the case studies presented below: N refers to the number of actors in the simulation, T refers to the number of waves we have in the panel study, *balance* refers to the observed marginals in the real datasets, allowing us to "cut" the continuous distribution to 0s and 1s, and *resolution* refers to the number of categories to simplify the latent variable, which is restricted to the binary case in this article.

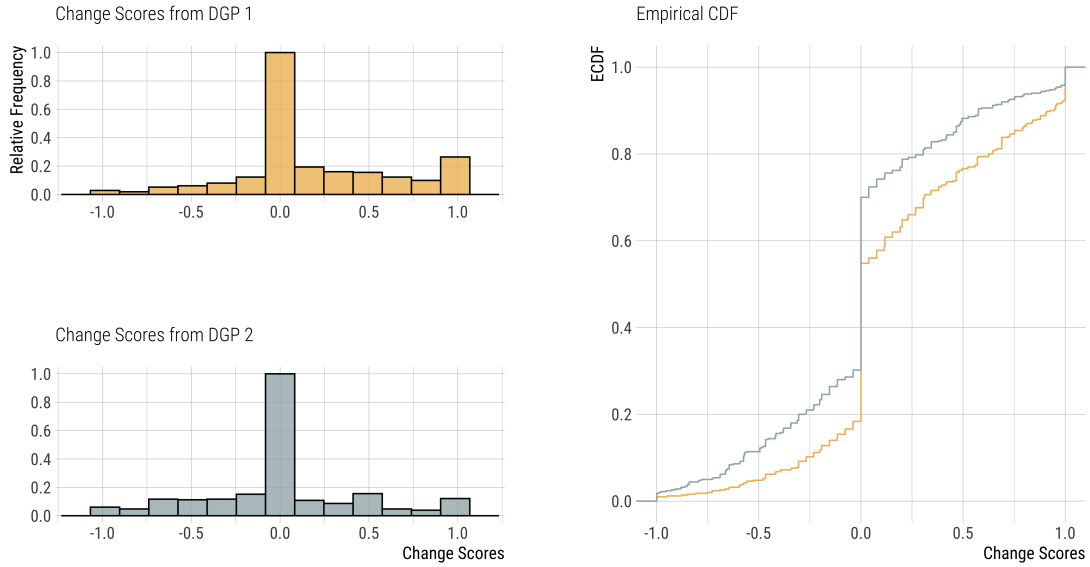


Figure 2: Calculating Distance for Observed and Simulated Trajectories

Notes: The figure presents an illustrative example of two known DGPs: (1) the top-left panel shows the distribution of change scores from DGP 1, (2) the bottom-left panel shows the distribution of change scores from DGP 2, and (3) the right panel shows the empirical cumulative distribution functions from each of these distributions.

How adequately does DGP 2 represent DGP 1? In the right panel of Figure 2, we show the empirical cumulative distribution functions (ECDFs) for our two distributions, which indicates that DGP 2 does a very poor job of reproducing DGP 1, with a Kolmogorov-Smirnov distance of 0.16. This means that we should be able to improve our predictions significantly if we start tweaking some parameters for DGP 2. Unfortunately, we of course do not know which parameters to tweak; the set of true change parameters is the exact thing we want to derive using observed trajectories. This leads us to consider a wide variety of parameter values that minimize the Kolmogorov-Smirnov distance scores between the observed trajectories and the said simulated trajectories.

Empirical Case Studies

In the following sections, we apply this procedure in different contexts, analyzing the *group-to-person generalizability* problem across real-world questions on political belief change.

In all case studies and simulation analyses, we restrict our attention to the stability and change of latent variables with binary outcomes to simplify our analyses.⁶ The replication files for the article are stored in <https://osf.io/5x48d/>. The R package for the use of this approach, “gridsearch,” can be accessed via the replication repo or <https://tkeskinturk.github.io/gridsearch/>.

⁶Since we use individual-level models, our approach is generalizable to responses with more than 2 categories. We restrict ourselves to the binary case to keep the mapping of latent scores to observed scores as smooth as possible, so that we do not have to impose additional assumptions that would detract us from the main point of our article.

Exploring the Case of Marijuana Legalization

Let us now return to the marijuana legalization case in the GSS and apply our grid-search procedure to see whether we can detect a set of plausible DGPs for our empirical data.

We first fit 1,000,000 simulations by drawing samples from four different parameters: *rate of change*, fixed between 0 and 1, *strength of change*, fixed between 0 SD and 2 SD, *directional balance of change*, fixed between 0 and 1, representing “all negative” change at 0 and “all positive” change at 1, and a *reliability score*, fixed between 60% and 100%. We use uniform priors for each parameter. We then calculate the Kolmogorov-Smirnov (KS) distance for each draw and keep KS distance values less than 0.005, ending up with 911 potential generative models.

Figure 3 shows the distribution of, and bivariate relationships among, these grid-search parameters, suggesting that the posterior distributions of strength and rate parameters are highly variable—and strongly correlated—while the posterior distributions for balance and reliability much more concentrated. Looking at the distributions, the interquartile ranges for strength and rate are 0.57 and 0.37, respectively, while these values are only 0.16 for balance and 0.06 for reliability.

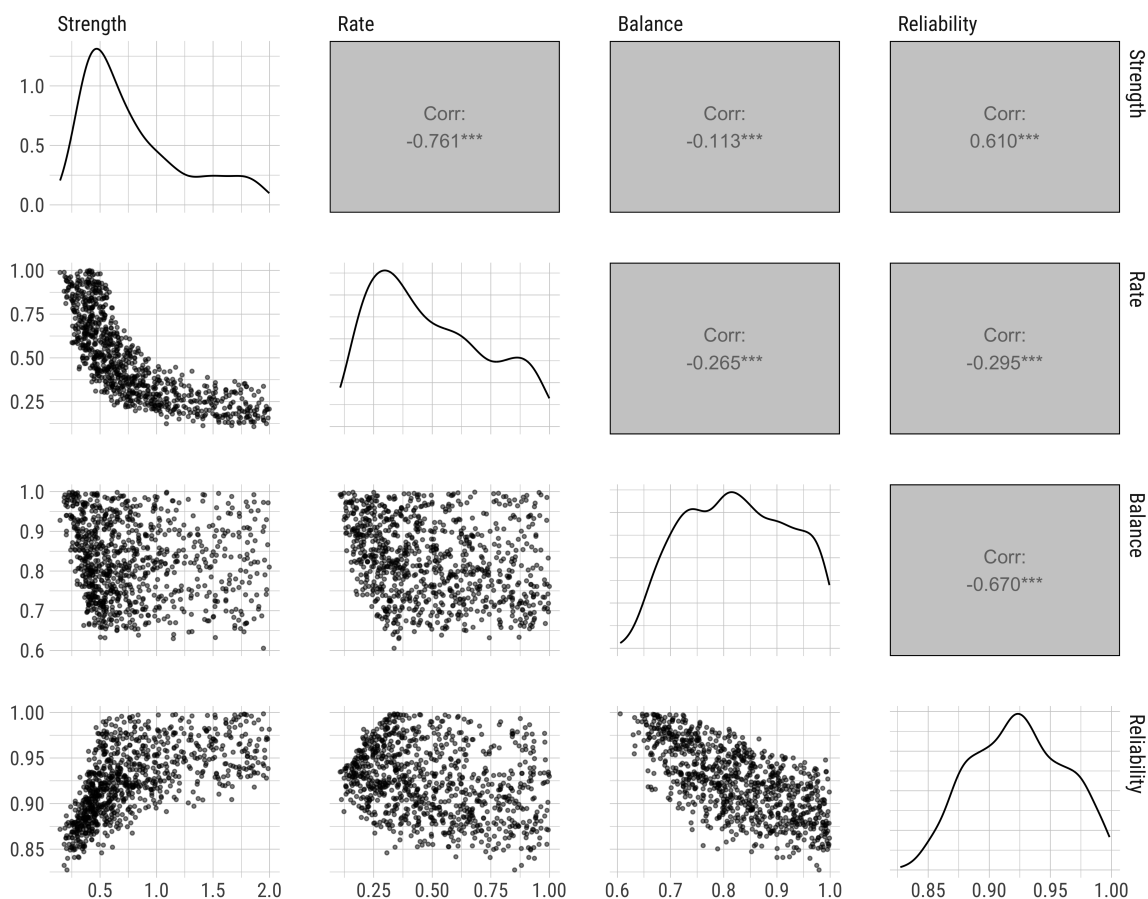


Figure 3: Pair Plots for Marijuana Legalization

Notes: The figure depicts the distribution of, and bivariate relationships among, grid-search parameter values from the best-fitting DGPs for the case of marijuana legalization (KS distance < 0.005).

These patterns make sense when we examine the conditional relationships among our parameters, presented in Figure 4. We see that the algorithm can easily suggest, at least for this particular case, a specific direction estimate—a median balance of 0.82 with an IQR of 0.75 and 0.90—but it converges to almost analytically determined bands in high-balance panels: we detect either small changes in a large number of people, or large changes in a small number of people. To put this result in perspective, note that the KS distance among these accepted draws are virtually indistinguishable from one another, most having a difference less than 0.001.

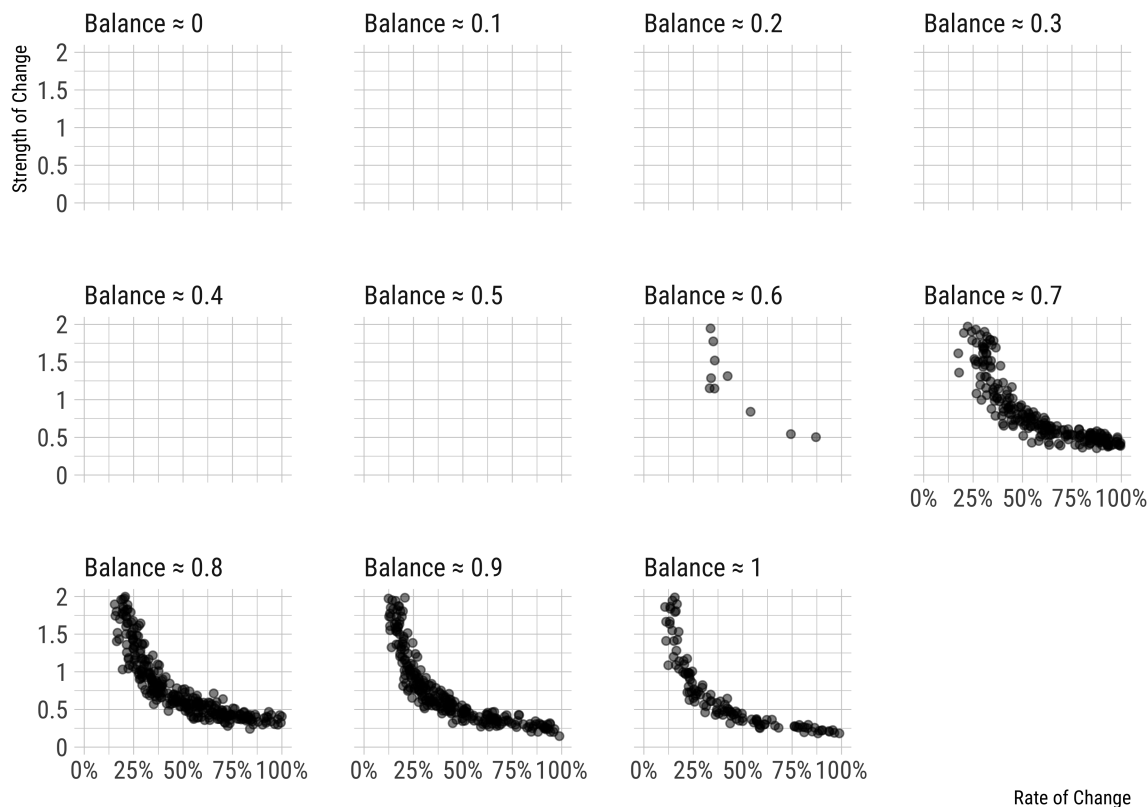


Figure 4: Best-Fitting DGPs for Marijuana Legalization

Notes: The figure depicts DGPs with KS distance < 0.005 (from 1,000,000 simulations) for the empirical case study. Each point represents an accepted draw. We rounded the balance parameters to show conditional relationships.

There is one strong implication of this exercise: there are competing generative models that may easily explain the same aggregate patterns, and there is no methodological trick to solve this issue. If we take the measurement error problem to its natural conclusions, the same data can be explained by incompatible theoretical mechanisms, and our conclusions may be wildly off-the-mark.

Exploring the Case of the Legality of Abortion

What about cases where the direction or even existence of change is less clear-cut? We now turn our attention to the legality of abortion debate, a contentious issue characterized by high disagreement and high political sorting in the American context (Baldassarri and Gelman 2008; Hout et al. 2022).

In 2006, 2008, and 2010 GSS panels, respondents agreed or disagreed with the statement: “it should be possible for a pregnant woman to obtain a legal abortion if the woman wants it for any reason.” Figure 5 shows response trajectories for this question across three panels, indicating a clear stasis.

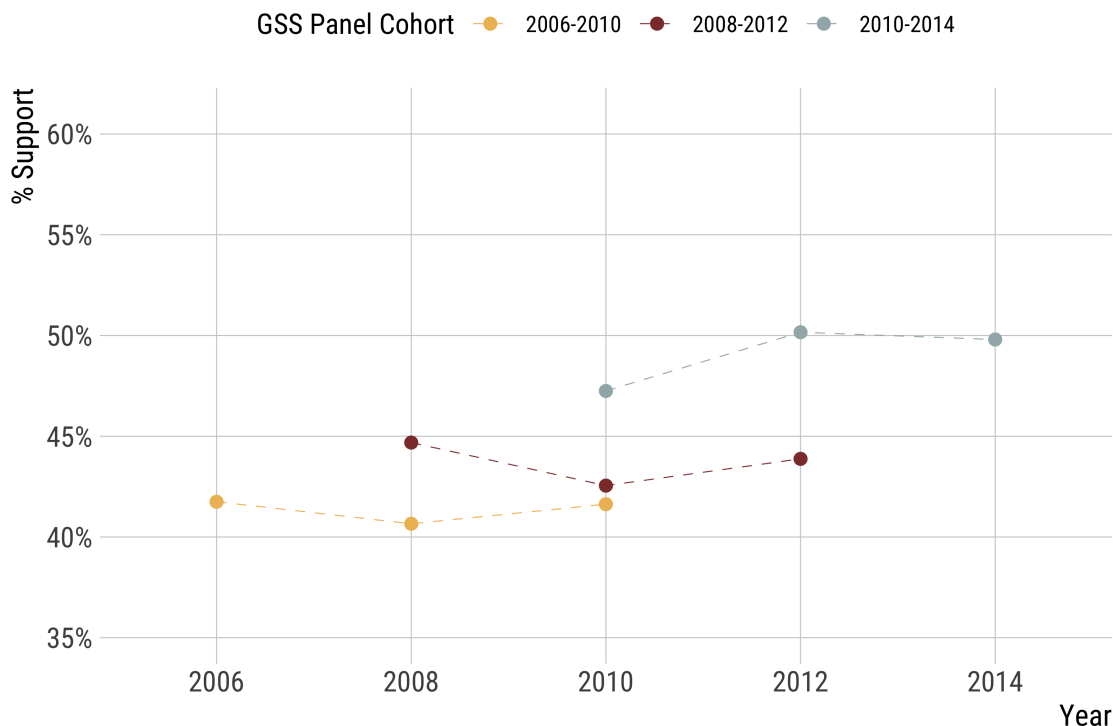


Figure 5: The Percent Support for the Legality of Abortion

Notes: The figure depicts the percent of American adults saying “it should be possible for a pregnant woman to obtain a legal abortion if the woman wants it for any reason” in the General Social Survey 2006, 2008, and 2010 panel studies. We included respondents who participated in all three waves of each panel study and weighted the results using GSS post-stratification weights. Note that panel attrition caused compositional changes in the estimates, and compared to the GSS time-series data, the estimates of support are marginally inflated.

We apply our grid-search algorithm once more, using the same specifications we applied in the case of marijuana legalization: 1,000,000 draws with $R \in [0, 1]$, $S \in [0, 2]$, and $D \in [0, 1]$, with reliability score ranging between 60% and 100%, each having uniform priors. This time, we kept DGPs with KS distance < 0.0025 given the high number of “better” fits we observed, resulting in 1,534 DGPs. Figure 6 shows the pair plots, while Figure 7 shows the conditional estimates.

We find strong variability in the plausible generative scenarios. Note how, in all cases, the accepted draws are squished towards 0—either via 0 rate of change or via 0 strength of change—indicating a clear preference towards no change in the underlying processes. That said, the plausibility bounds also become strongly noisy when the estimated balance is even: accepting scenarios ranging from no change to the now-familiar high change in a low number of people and low change in a high number of people. The IQR of KS distance among all accepted samples is between 0.001 and 0.002, suggesting that model fits are reasonably similar across all scenarios.

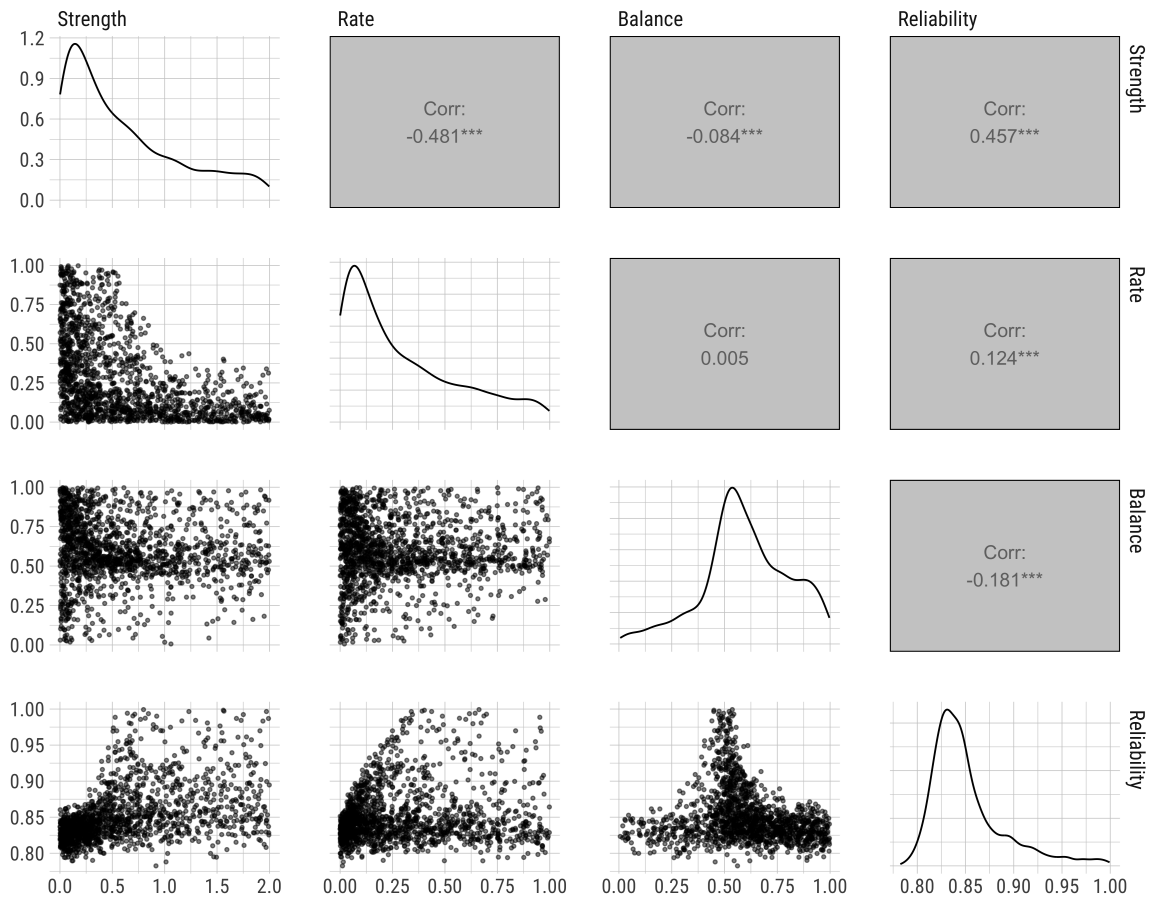


Figure 6: Pair Plots for the Legality of Abortion

Notes: The figure depicts the distribution of, and bivariate relationships among, grid-search parameter values from the best-fitting DGPs for the case of the legality of abortion (KS distance < 0.0025).

This exercise is an appropriate illustration of the limitations of panel studies. Based on the observed data, we cannot determine whether the observed marginals result from a small number of people making substantial belief changes or a large number of people making minor political changes. This may, ultimately, boil down to the issue of observation frequency: with three-wave panel data and reasonable doubts about the accuracy of every single survey response due to reliability and resolution concerns, it is challenging to adjudicate the underlying generative models. In the next section, we relax this assumption and extend our method to cases with more than three waves.

Exploring the Case of Immigration Spending

The data for this exercise comes from the *Political Psychology Data from a 26-Wave Yearlong Longitudinal Study* (Brandt, Turner-Zwinkels, and Kubin 2021), conducted biweekly in 2019-2020 for a year. In one question in the survey, respondents were asked whether they agree with the policy of increased federal spending on immigration control. We binarized the responses to this question such that those agreeing with the statement received a 1, and those who said the spending should

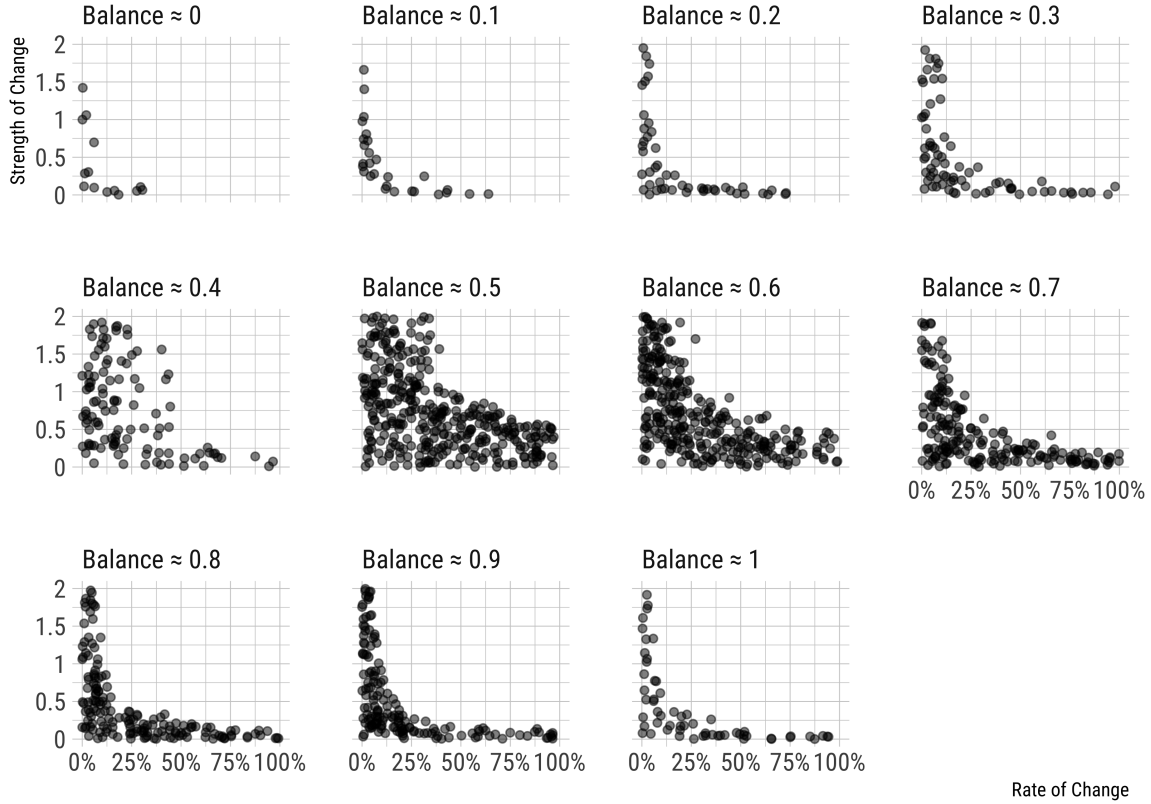


Figure 7: Best-Fitting DGPs for the Legality of Abortion

Notes: The figure depicts DGPs with KS distance < 0.0025 (from 1,000,000 simulations) for the empirical case study. Each point represents an accepted draw. We rounded the balance parameters to show conditional relationships.

be decreased or kept the same received a 0. We then dropped respondents with missing data and trimmed the periods to pre-COVID windows, resulting in 213 individuals observed 17 times over 34 weeks. Although the sample for this case study is obviously not representative, it provides an intensive set of repeated observations at the individual level (see Figure 8).

We applied the same procedure, but we found that our KS distance cutoff of 0.005 was too low—we ended up with no accepted samples, suggesting that the model simply cannot converge to a reasonably strong fit for this case study (we discuss the implications of this finding below). We then redefined the cutoff point as a KS distance of 0.05, ending up with 3,676 accepted samples.

Once again, Figure 9 presents the pair plots, while Figure 10 depicts the conditional DGP estimates. Consistent with the decline in the observed marginals, we find accepted samples in the regions with balance lower than 0.5 (a median of 0.14 with an interquartile range between 0.07 and 0.22). We also find similar IQR values of 0.35 for strength and 0.37 for rate, suggesting plausibility bounds in our conditional estimates. In contrast to marijuana legalization case, however, these plausibility bounds have much more noise, as seen in Figure 10, suggesting that the data has many plausible DGPs spread across the grid.

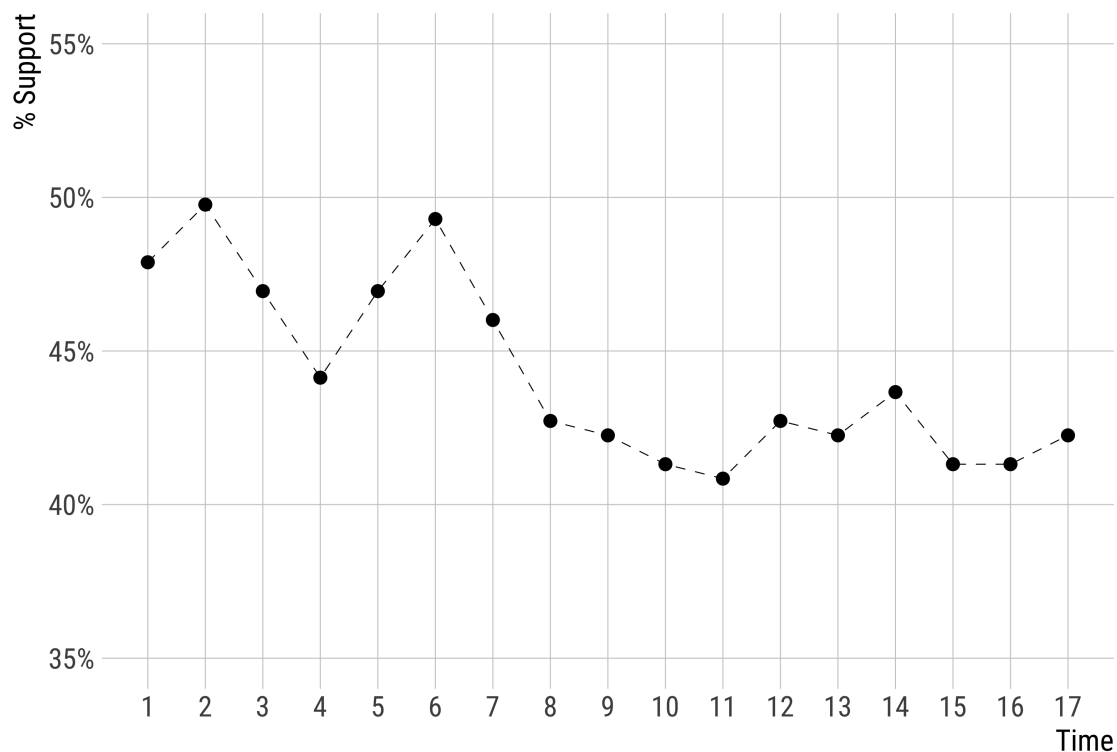


Figure 8: The Percent Support for an Increase in the Control of Immigration Spending

Notes: The figure depicts the percent support for the statement "the federal spending to control immigration should be increased" in the Political Psychology Data from a 26-Wave Yearlong Longitudinal Study.

The Detection of True Changers

So far, we have presented evidence for identifying plausible DGPs that may have produced the observed empirical data. But what about classifying individuals into those who change and those who do not, i.e., knowing *who changed*? Can we also recover an accurate classification of changers and non-changers at the individual level of observation (Tharwat 2020)?

To address this question, we conduct a simulation exercise. For each of our empirical case studies, we first generate 1,000 datasets by fixing the parameters of interest to observed and assumed values. For the case of marijuana legalization, for instance, we fix the number of individuals, the number of waves and the balance of the marginals to the observed GSS values, while we fix the balance and reliability scores to the median of our grid-search results, and rate of change and strength of change to a plausible scenario for 50% and 1 SD, respectively. Similarly, for the legality of abortion and immigration spending, we assumed 20% rate with 2 SD strength as well as 100% rate with 0.25 SD strength, while fixing everything else to observed values in the data or median values in our estimates. These scenarios are reasonable fits to our cases, but for each of 1,000 simulated datasets, we *know* who the real changers are.

Our metric for classification success is quite straightforward. We first use the "observed" simulation data to fit individual-level regression models:

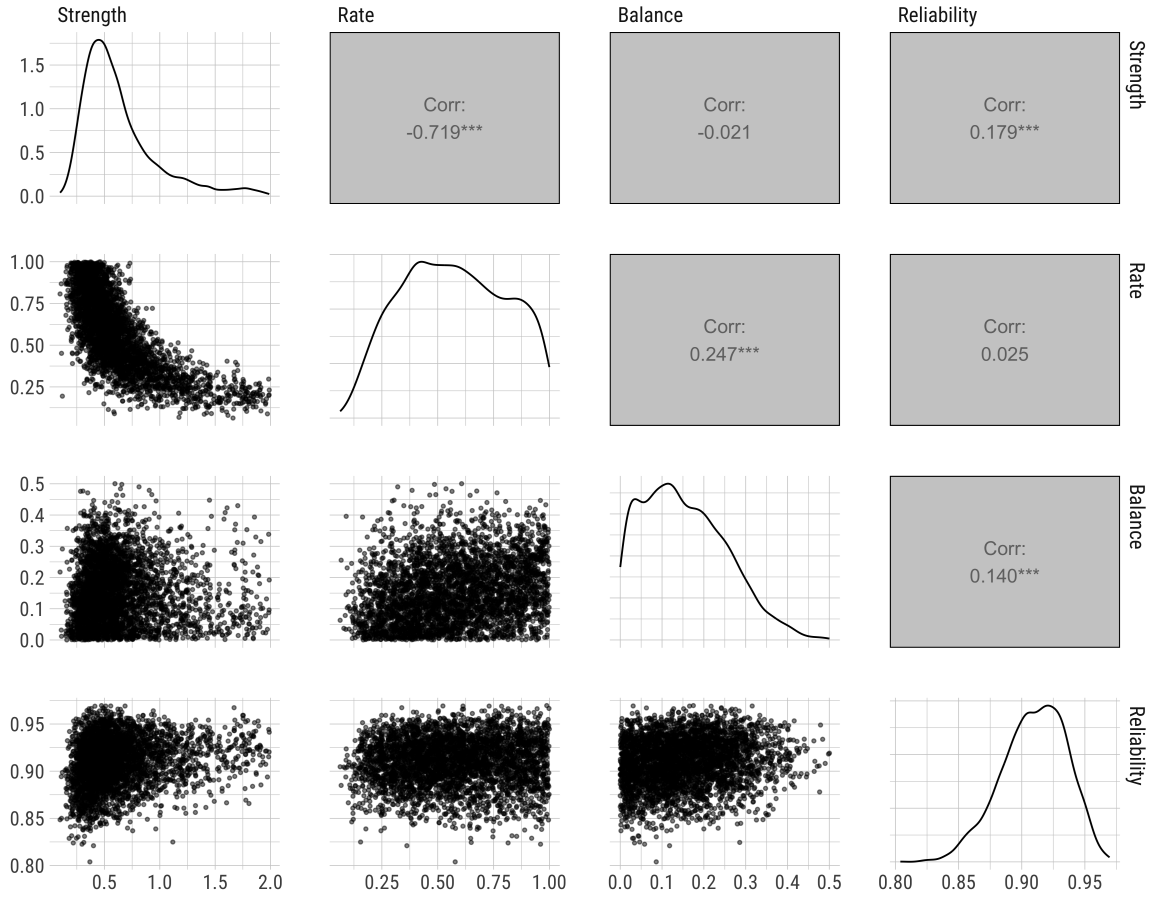


Figure 9: Pair Plots for Immigration Spending

Notes: The figure depicts the distribution of, and bivariate relationships among, grid-search parameter values from the best-fitting DGPs for the case of immigration spending (KS distance < 0.05).

$$\mathbb{P}(y = 1) = \beta_0 + \beta_1 t, \quad t \in [0, 1]$$

where t is a normalized time variable. We then estimate a predicted change score by calculating the difference between the prediction at $t = 1$ and the prediction at $t = 0$.

Once we have all the change scores, we rank individual cases from highest to lowest, and based on our true DGP parameters, select respondents as “changers.” For instance, if we assumed 70% positive change and 30% negative change, we select the highest 70% of individual slopes as positive changers and the lowest 30% of individual slopes as negative changers. In the last step, we examine whether we can recover true changers in our DGPs from these observed changers.

Figure 11 shows the distribution of accuracy scores across all studies, each having 1,000 simulations for change detection. The median accuracy score is 43% for the case of marijuana legalization, 36% for the legality of abortion, and 61% for immigration spending, meaning that among all true changers, we can only correctly classify 36-61% of them, even with the full knowledge of the DGPs.

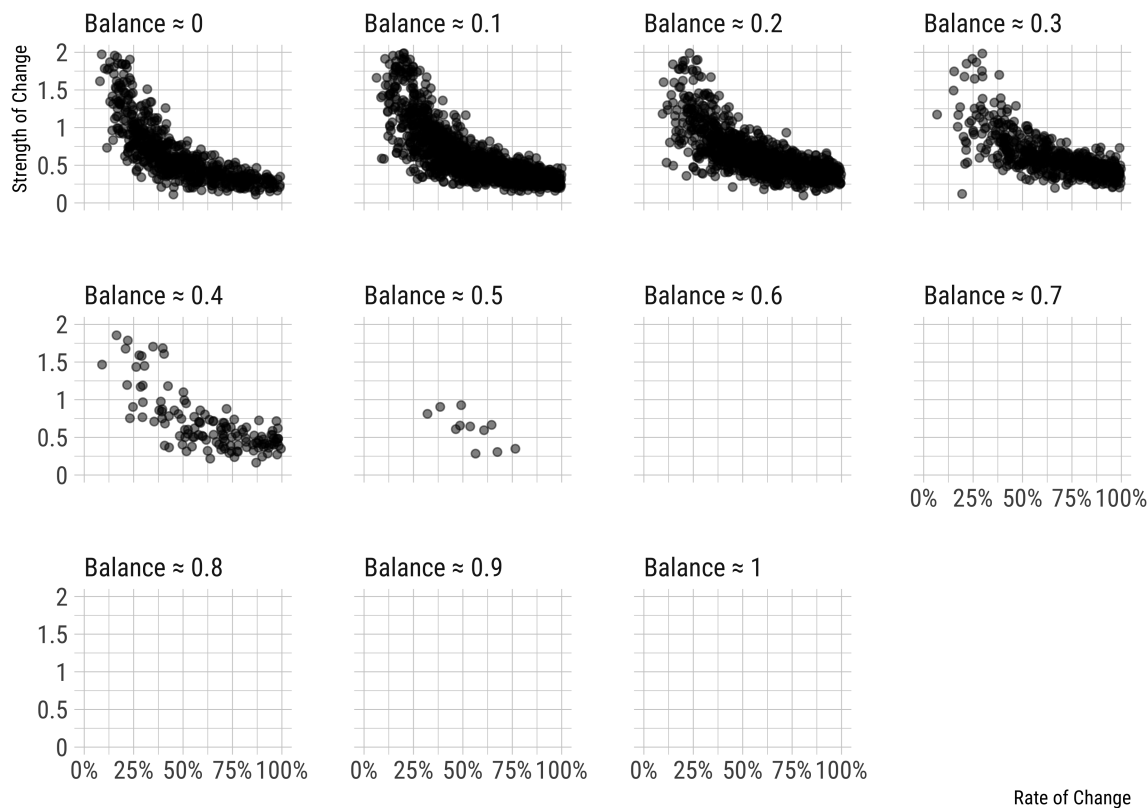


Figure 10: Best-Fitting DGPs for Immigration Spending

Notes: The figure depicts DGPs with KS distance < 0.05 (from 1,000,000 simulations) for the empirical case study. Each point represents an accepted draw. We rounded the balance parameters to show conditional relationships.

Theoretical and Empirical Implications

We now examine the main conclusions of our case studies, and draw several implications for the use of panel data in studies of political change and stability across the life-course.

1. Our case studies highlight that the aggregate change patterns we observe may emerge from vastly different compositional processes. This implies that researchers should be cautious when interpreting the results from panel studies, and think carefully about the measurement problem. To bolster this point, we provide two additional studies in Supplemental Materials A—the first case looking at a tumultuous social period during the COVID-19 pandemic and examining patterns of “generalized trust” (using data from the British Election Study panel), and the second case looking at adolescents over a 10-year window to explore “belief in God” (using data from the National Study of Youth and Religion).
2. We emphasize that this identification problem cannot be “solved” with simple methodological considerations. As we emphasized before, our objective was not to eliminate the problem, but rather to provide a concrete approach that might help us understand the issue more clearly. More substantively, we need to acknowledge the problem in our empirical applications and

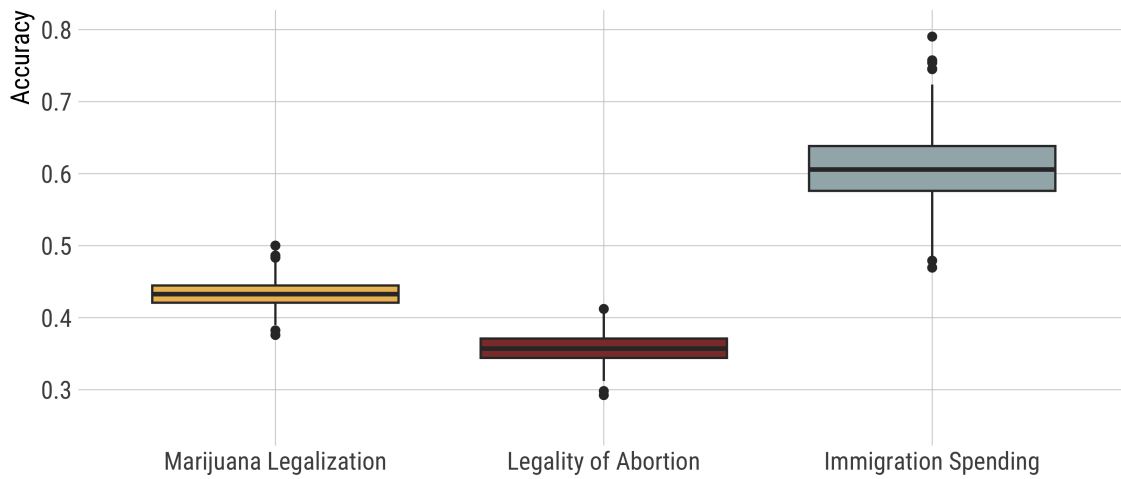


Figure 11: The Distribution of Accuracy Scores Across Empirical Case Studies

Notes: The figure depicts the distribution of accuracy scores for the detection of changers across three empirical case studies. For each case, we performed 1,000 simulations for the classification task.

be willing to *impose* certain assumptions. The fact that multiple—sometimes contradictory—generative processes may very well account for the same data means that researchers should assert theoretical positions when it comes to connecting their hypotheses to empirical data.

3. Our studies present interesting variations across (a) model fit, i.e., whether our grid-search approach can get reasonably close to true DGPs, and (b) the noise around plausibility bounds. To understand these dynamics better, we conducted a simulation exercise and explored the effects of reliability and the number of waves on model fit (see Supplemental Materials B). We show that an increased reliability score and higher number of waves mean increased model fit—the effects of which depend on the underlying real-world parameters. This suggests that a better fit for plausible DGPs requires researchers to collect as many waves as possible and a strong investment in reliable response measurement strategies. Without these, researchers will need strong theoretical assumptions to narrow down plausible data generating mechanisms.
4. We showed that accurately identifying changers at the individual level might be challenging. In Supplemental Materials C, we present findings from a simulation study, which explores 360 DGPs across a wide variety of scenarios—varying rate, strength, directional balance, reliability, and the number of waves. We show that accuracy scores range from 8% to 74%, with more reliability and higher number of waves significantly increasing our ability to classify individuals correctly. Once again, these findings suggest that researchers should strive to increase the reliability of response measurement and try to collect as many study waves as possible in order to have a decent shot at identifying change at the individual level.⁷

⁷In Supplemental Materials D, we show additional analyses comparing our fixed-effects classification approach with multilevel mixed models (Lersch 2023). We found that a fixed-effects approach largely fares better, even though a mixed model incorporates information from the overall sample distributions. This, however, largely reflects the fact that we have not incorporated a mixture process: a mixture mixed effects model or a latent transition model might have allowed us to improve these estimates. Since, for the purposes of this paper, we focused on the group-to-person generalizability

Discussion and Conclusions

In this article, we examined the identification of belief change in panel context. Using simulations and empirical case studies, we showed that a grid-search procedure that iterates over generative models can help us identify plausible data generation processes that may have produced our empirical observations, though the identification of individual cases occurs under narrow conditions. We argued that the *group-to-person generalizability problem* is an impediment to a robust understanding of political belief change, and scholars should understand the extent to which their target estimates may generalize to the population of their study. As such, we proposed a simulation approach that takes response generation seriously, which may help us move forward in addressing this problem.

This article has several implications for the study of political belief change across the life-course. First, our findings suggest that recent debates about the existence or non-existence of belief change in adult populations (Kiley and Vaisey 2020; Lersch 2023) should tackle the heterogeneous nature of change and stability in individual preferences because group-level measures might hamper our understanding of individual mechanisms. Instead, scholars need to specify the generative models that aggregate average estimates imply. Just like analyses that examine population patterns to adjudicate compositional and individual changes (Bartels and Jackman 2014; Vaisey and Lizardo 2016), we need robust and reliable panel studies to identify the underlying generative processes.

Second, the ambivalence in response trajectories indicate that classical solutions to the measurement problem—ranging from scale construction to structural equation models and principal component analyses (Alwin and Krosnick 1991; Ansolabehere, Rodden, and Snyder 2008; Judd and Milburn 1980)—should have an important place in panel designs, as even moderate reliability scores can significantly hinder our ability to specify plausible generative processes.

One theoretical assumption of this article was to rely on a latent variable model, and avoid questions about contradictory considerations in human cognition that mitigate stable and coherent opinion formation (Converse 1964; Zaller and Feldman 1992). This issue, of course, introduces another layer to the problem of belief change and measurement error. We relied on a very strong assumption that an individual has a central tendency when it comes to political issues, rather than being functionally “random” in their responses to survey questions. However, a careful examination of adult belief change should be grounded in a more theoretically-sound study of belief formation.

Finally, these results call for granular panel surveys that go beyond the classical 2-wave and 3-wave designs because reliability concerns significantly limit the plausible conclusions we may derive from these shorter panels. In a 2-wave setup, where there is only one change, we can never properly establish whether a change we observe is due to measurement error or it is indeed a genuine shift, making extended panel designs necessary. Since we care about generative models rather than one observed sample, strategies that enable us to reliably estimate target quantities are warranted and essential to make progress on longstanding questions about belief change.

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problem, we bracket this issue from further consideration, but we hope future research will improve this strategy.

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Supplemental Materials for

The Promises and Pitfalls of Using Panel Data to Understand Individual Belief Change

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January 2025

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In this document, we provide supplemental materials for the article, *The Promises and Pitfalls of Using Panel Data to Understand Individual Belief Change*.

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1 Supplemental Materials A

We present findings from two additional case studies to showcase our approach—the first looking at generalized trust and the second looking at belief in God.

1.1 The Case of Generalized Trust

The first empirical example comes from the British Election Study panel, which, between the years of 2019 and 2024, asked its panel respondents whether “most people can be trusted, or that you can’t be too careful in dealing with people?” We show the percent agreement with “most people can be trusted” in Figure A1, which describes the trajectory across 4,116 individuals over 5 waves, in a period with tumultuous social problems during the COVID-19 pandemic.

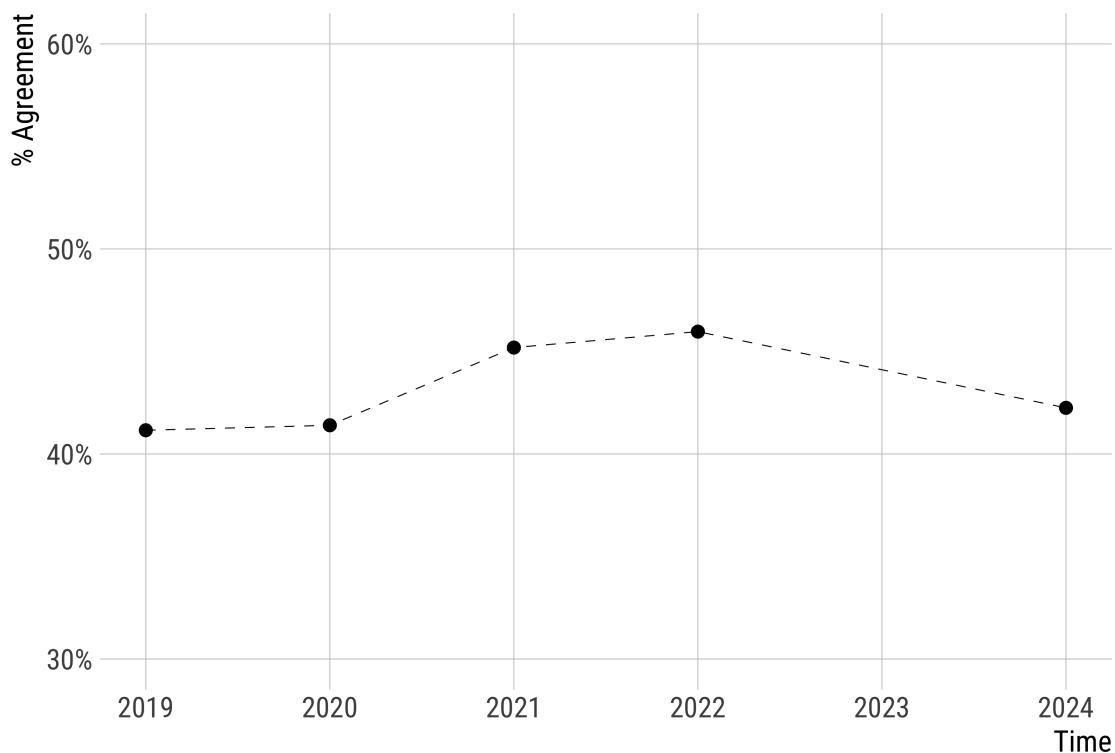


Figure A1: The Percent Agreement with Generalized Trust

Notes: The figure depicts the percent support for the statement “most people can be trusted” in the British Election Study panel data. We included respondents who participated in all five waves of the panel study.

Figure A2 presents the pair plots, while Figure A3 depicts the conditional DGP estimates. We see the familiar plausibility bounds on the bottom left quadrant, though we find that the estimates are much less noisy. The balance parameter is localized in a median value of 0.77 (with an interquartile range of 0.69 and 0.88), suggesting a small increase in generalized trust in the period, though, once again, we see plausible scenarios with large changes in a small number of people or small changes in a large number of people. This finding is significant: even with a consequential social event, our identification problem persists regardless of panel length or exogenous shocks.

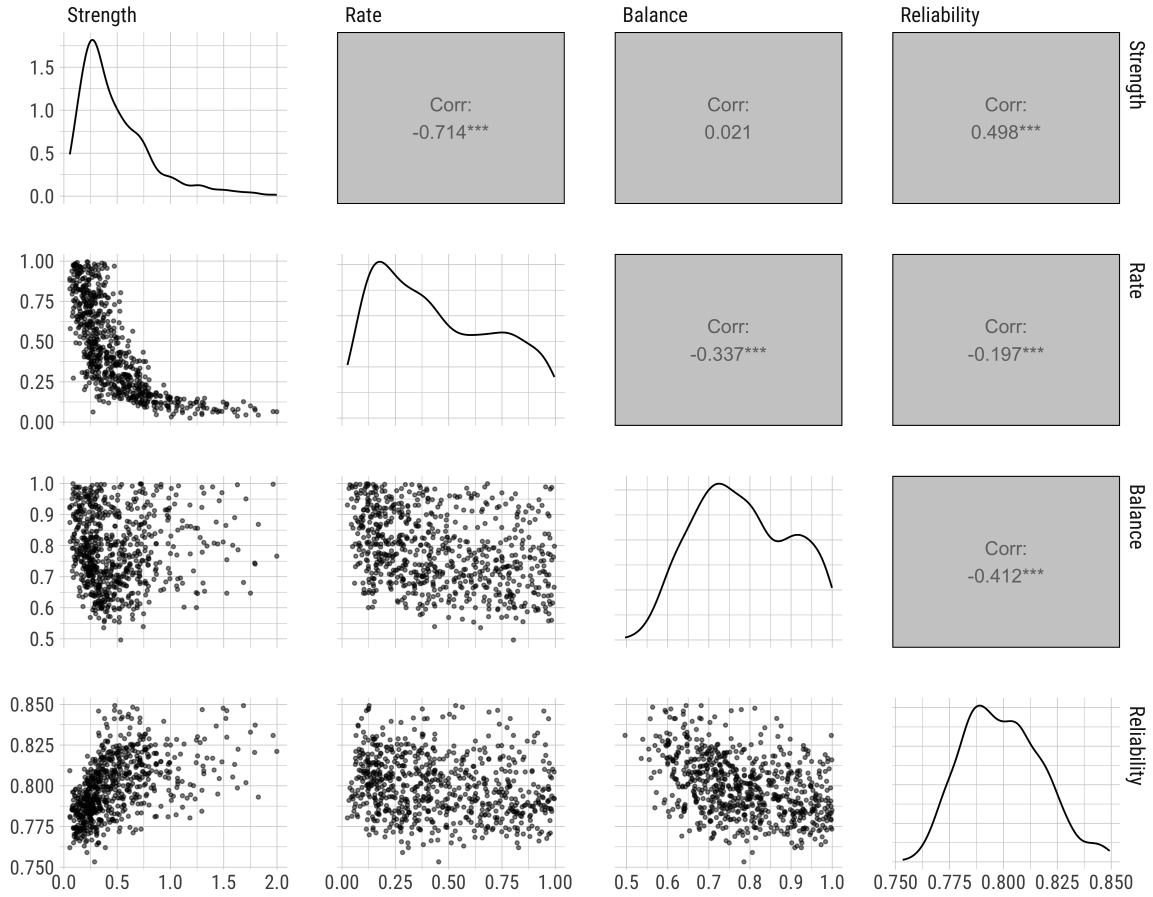


Figure A2: Pair Plots for Generalized Trust

Notes: The figure depicts the distribution of, and bivariate relationships among, grid-search parameter values from the best-fitting DGPs for the case of generalized trust (KS distance < 0.01).

1.2 The Case of Belief in God

For our second empirical example, we turn to the National Study of Youth and Religion (NSYR), a nationally representative study of U.S. adolescents between the ages of 13 and 17, followed over a 10-year period. The NSYR asked respondents, across four waves, whether they believe in God. We coded the responses such that those who stated they “certainly” believe in God received a 1, while those stating “maybe” or “no” received a 0. Figure A4 describes the trajectory of 1,728 adolescents between the years of 2003 and 2013—a period where most individuals go through substantive changes in their cultural and political beliefs and preferences (Kiley and Vaisey 2020).

The results are presented in Figures A5 and A6. This time, we see much tighter bands closer to the top-right quadrant: the algorithm detects substantial changes reflected in the plausibility bounds. We see a median value of 0.31 (an interquartile range between 0.21 and 0.38) for directional balance, suggesting loss of belief when respondents move from adolescence to early adulthood. Once again, the expected line is a combination of competing DGPs; however, the range is smaller for the rate of change—with an IQR of 0.26—simply because the bounds are tighter on the top end.

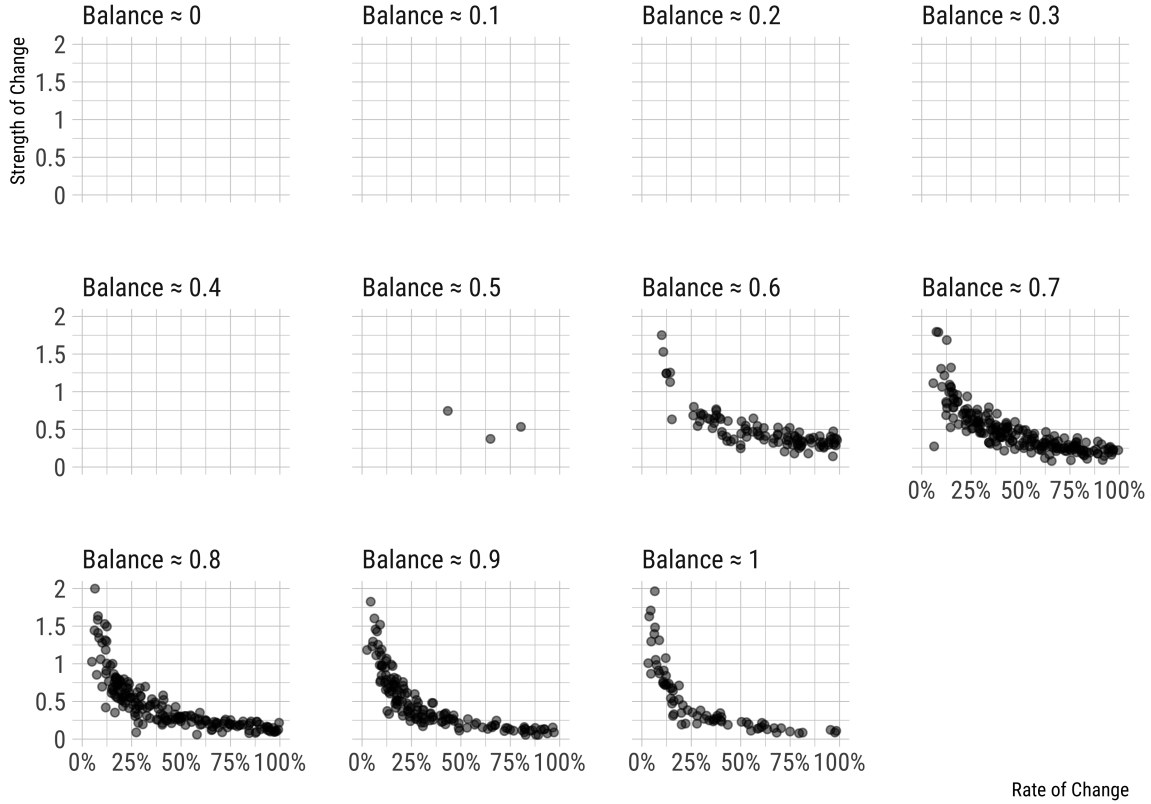


Figure A3: Best-Fitting DGPs for Generalized Trust

Notes: The figure depicts DGPs with KS distance < 0.01 (from 1,000,000 simulations) for the empirical case study. Each point represents an accepted draw. We rounded the balance parameters to show conditional relationships.

2 Supplemental Materials B

In our case studies, we notice that certain plausibility bounds are much more clustered than others. For instance, the plausibility bounds for “generalized trust” have much tighter bounds than the bounds for, say, “the legality of abortion.” This practically gives us a sense of “fit”—our algorithm is sometimes better to get a sense of what an analytical solution line looks like than other times. In what follows, we show simulation findings that suggest that this model fit is a function of the number of waves a study has and the reliability of response measurement.

We first generate 96 datasets by varying the number of waves (3, 5, 7, and 10), the rate of change (10%, 25%, and 50%), directional balance (polarization and secular change) and reliability (60%, 70%, 80%, and 90%). We fix the number of individuals the studies have to 10,000, the marginal balance to 50%, and the strength parameter to 1 SD. For each of these datasets, we then fit our grid search procedure 100,000 times, and subset the parameter combinations with a KS distance less than 0.05. This provides us with a set of plausibility bounds to check model fit.

To do so, we fit a log-log model using the following specification:

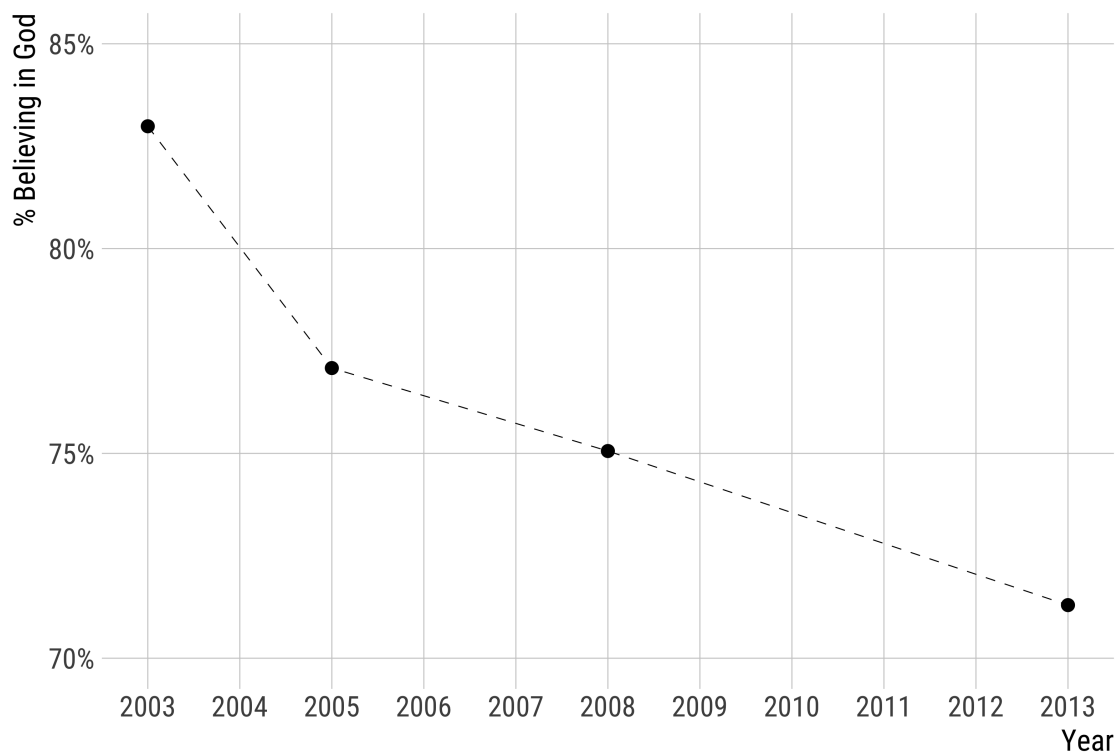


Figure A4: The Percent Beliving in God

Notes: The figure depicts the percent reporting that they certainly believe in God in the National Study of Youth and Religion panel data. We included respondents who participated in all four waves of the panel study.

$$\log(\text{Rate of Change}) = \beta_0 + \beta_1 \cdot \log(\text{Strength of Change}) + \epsilon$$

after which we calculate the R-Squared (R^2) value of the model fit. In the end, we end up with 96 R^2 values representing the model fit across different parameter combinations.

Figure B1 shows the R^2 values across different parameter combinations. There are several important findings. First, we see that having higher number of waves usually increases the model fit, much more so for cases where there is a secular change trend and at least 25% rate of change. When we fit a linear regression model using these values, we see that having 10 waves instead of 3 increases our model R^2 by 15%, net of all other parameters. Second, we find that, similarly, more reliability means better fit: increasing response reliability from 60% to 90% means 10% increase in R^2 values. In general, the best R^2 values emerge when we have at least 7 waves and 90% reliability score, even though the underlying generative models are the same for all adjacent cases.

The implications of this simulation exercise are straightforward:

1. There are things we cannot change, which would allow us to have a better chance at detecting plausible DGPs, such as the rate of change or the direction of change.

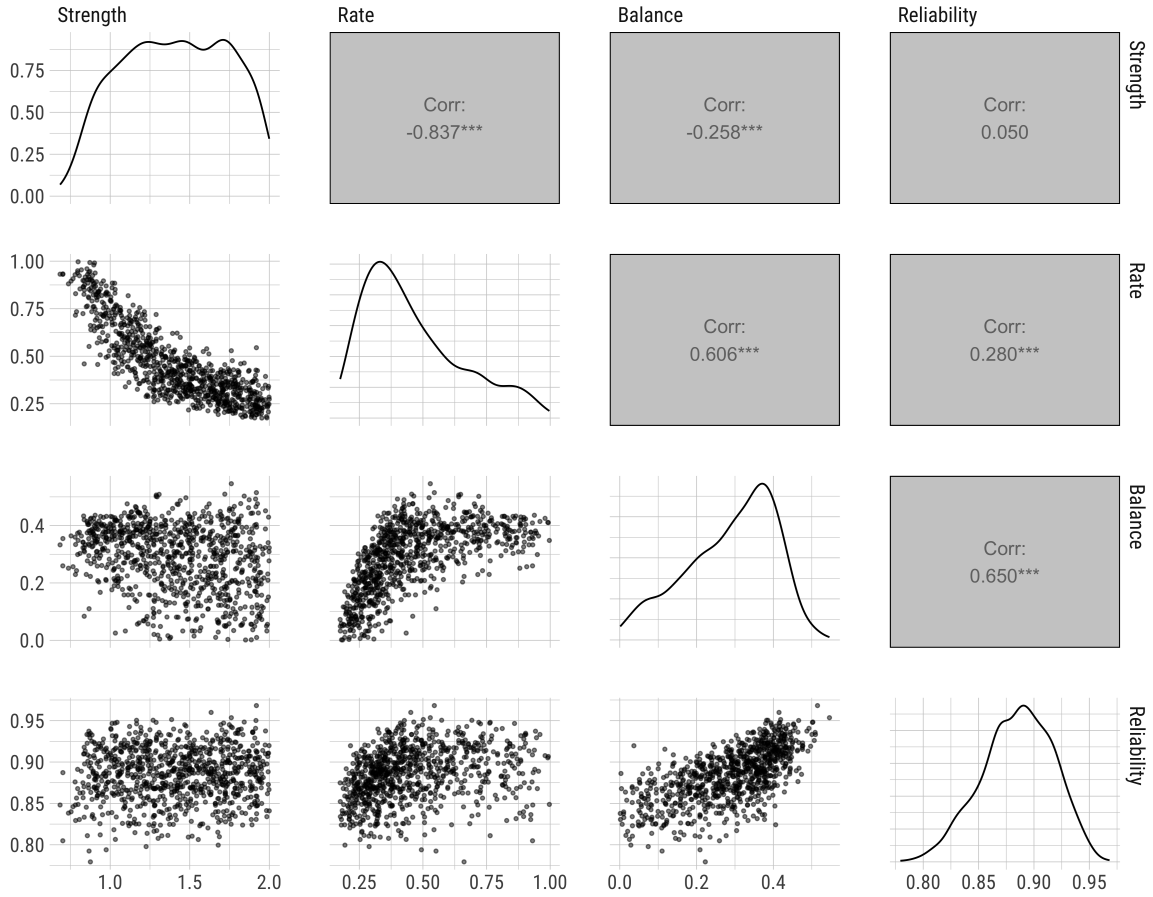


Figure A5: Pair Plots for Belief in God

Notes: The figure depicts the distribution of, and bivariate relationships among, grid-search parameter values from the best-fitting DGPs for the case of belief in God (KS distance < 0.01).

2. There are, however, things within our reach that we need to manipulate: researchers should design studies with as many waves as possible to get a better shot at identifying the set of plausible DGPs, and they need to use reliable response measurement strategies to distinguish true change from measurement error.
3. Finally, researchers should note that while these strategies would allow us to have better fit—defined as less noise around the analytical solution—they do not “solve” the identification problem; rather, they give us a more precise solution line for our main estimates.

In the next section, we look at the question of whether identifying changers from non-changers has similar requirements in terms of data collection.

3 Supplemental Materials C

In our empirical case studies, we showed that even with the full knowledge of the DGPs underlying any given data, accurately identifying changers at the individual level can be quite challenging. We

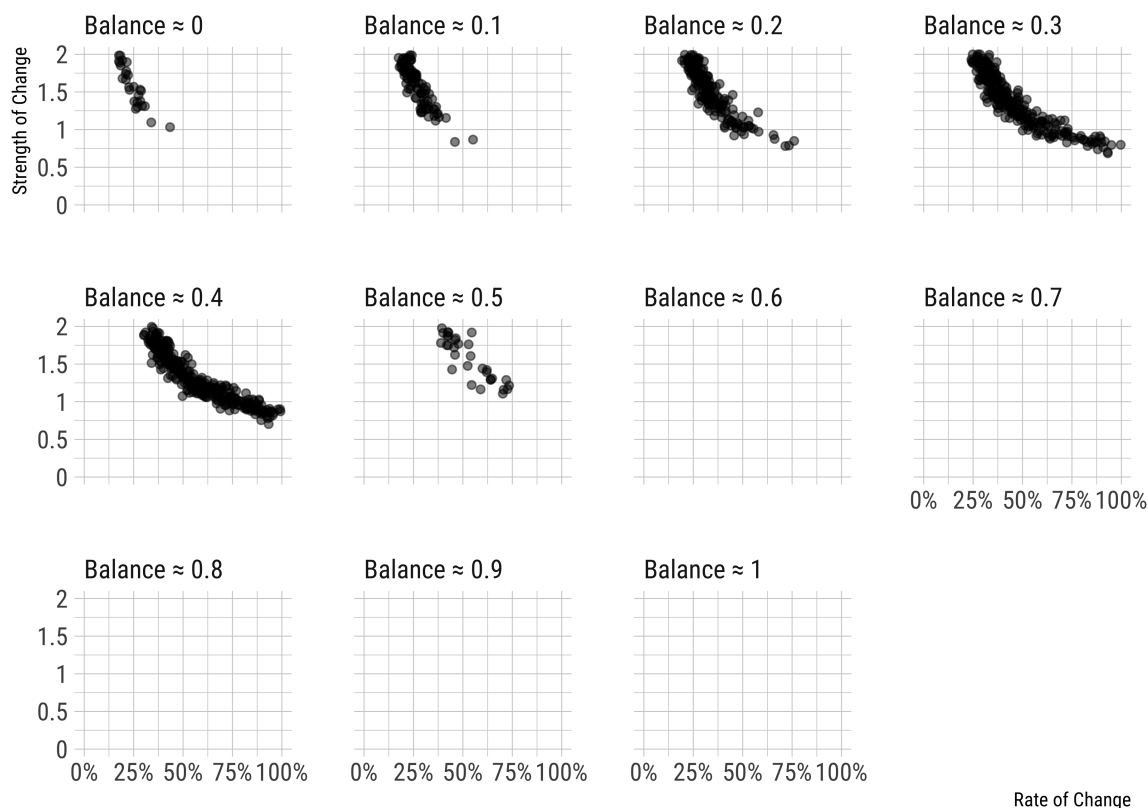


Figure A6: Best-Fitting DGPs for Belief in God

Notes: The figure depicts DGPs with KS distance < 0.01 (from 1,000,000 simulations) for the empirical case study. Each point represents an accepted draw. We rounded the balance parameters to show conditional relationships.

now ask under what conditions this identification works better.

To address this question, we conduct another simulation exercise. We first generate 360 datasets by varying several parameters of interest in the DGPs: 10,000 individuals across various number of waves (3, 5, 7, and 10), rate of change (10%, 25%, and 50%), strength of change (0.5 SD, 1 SD, and 2 SD), direction of change (polarization and secular change), and reliability of change (60%, 70%, 80%, 90%, and 100%). For each of these conditions, we generate 1,000 simulated data (ending up with 360,000 datasets) where we know, by our design, who changed throughout the observation window. We then apply our procedure of classification described in the article.

We found a median of 0.39 accuracy rate from these simulations (with an interquartile range of 0.28 and 0.53), meaning that among all true changers, we can only correctly classify 39% of them at the median, even with the full knowledge of the DGPs. Therefore, it is clear that our general ability to identify true changers, at least at the median, is quite low.

Under what conditions could we do a better job in classifying true changers? Researchers cannot influence rate of change, strength of change, or the direction of change, but they may modify the reliability score of their measure—e.g., by asking several questions to increase item reliability—or

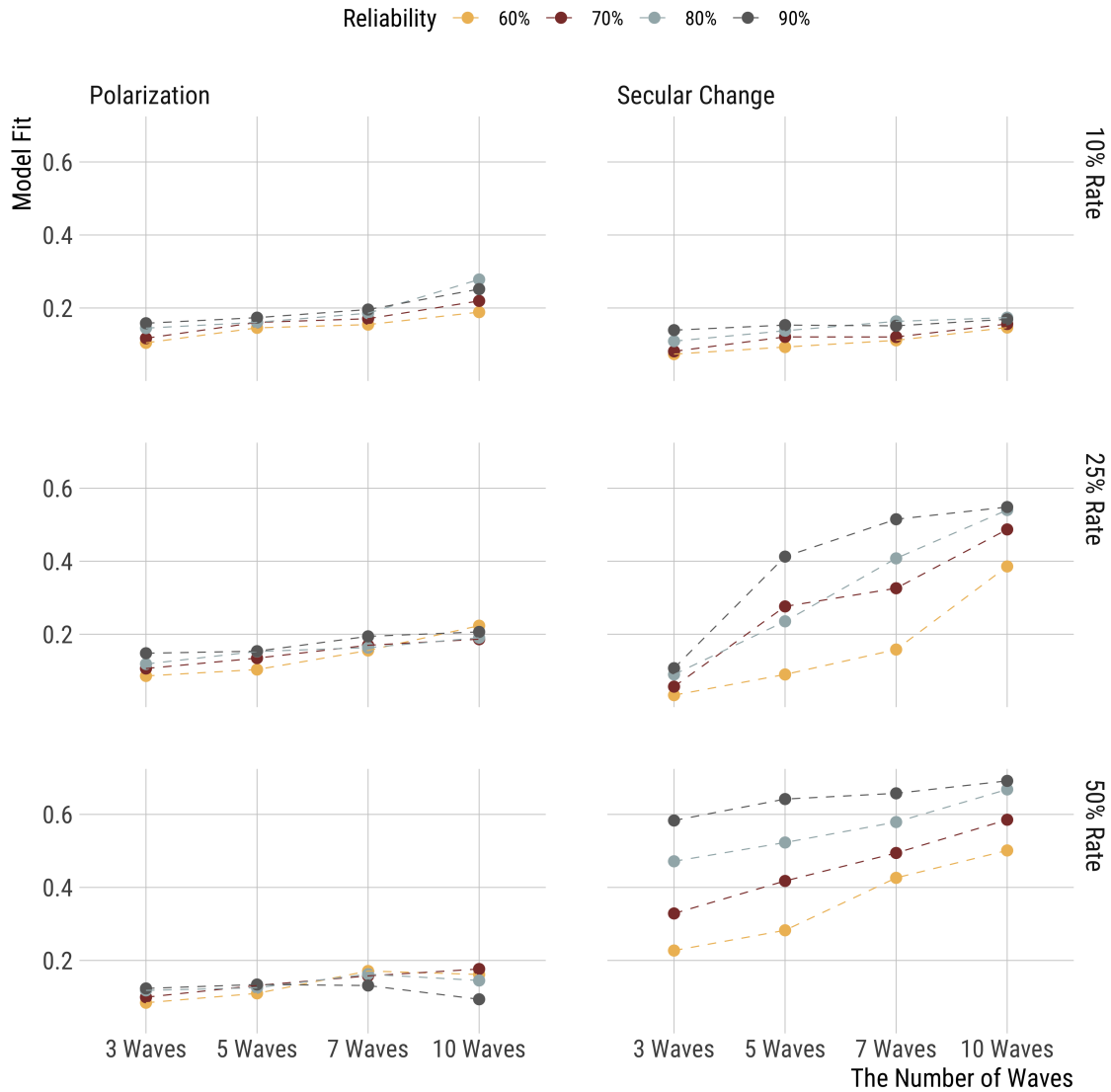


Figure B1: Reliability and Number of Waves on Model Fit

Notes: The figure depicts the model fit values across reliability score and the number of waves.

the number of waves in a panel study. In Figure C1, we examine how changing reliability and the number of waves affects our ability to detect true changers. Strikingly, we find a monotonic and almost linear effect of reliability and the number of waves on accuracy. To put it bluntly, we need more waves and reliable measures to increase our chances of proper detection of change.

These findings suggest that, while the grid-search procedure might allow us to identify plausible generative models that may have generated the empirical patterns, identifying true change at the level of individuals occurs under only very narrow conditions. The finding that simulations with longer panels performed better, on average, than the three-wave cases indicates that having longer panels might be essential to dampen the negative effects of response reliability on our estimates.

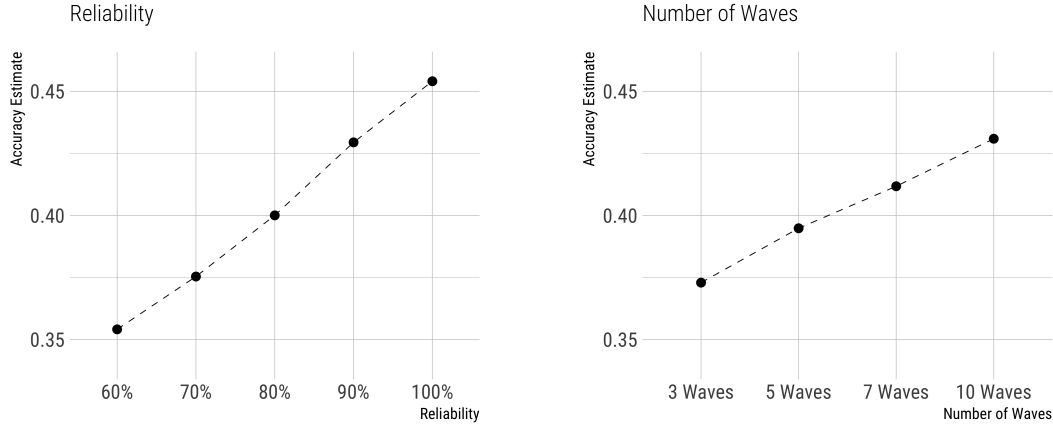


Figure C1: Reliability and Number of Waves on Classification Accuracy

Notes: The figure depicts average marginal predictions of reliability score and the number of waves on accuracy.

This finding emphasizes the importance of collecting panel data over more extended periods, as well as using measurement models that explicitly aim for higher reliability.

4 Supplemental Materials D

We presented evidence showing that identifying “true change” at the level of individuals occurs under only very narrow conditions. That said, one assumption we made while doing these analyses was to employ a fixed effects approach—estimating change scores at the individual level—without relying on population parameters that would allow us to have more plausible values (Lersch 2023). In this alternative approach, we could achieve the calculation of change scores by fitting the model

$$y_{it} = \beta_0 + \alpha_{0i} + (\beta_1 + \alpha_{1i})t_{it} + \epsilon_{it}$$

where t is normalized between 0 and 1, and individuals has varying intercepts and slopes.

Using this mixed model, we may simply apply the same ranking procedure we implemented in the main manuscript, but the individual-level change scores would be computed by partial-pooling, allowing us to *smooth* change distributions in identifying overall trajectories.

In Figure D1, we present one such simulation exercise across 32 DGPs where we vary the number of waves (3 and 10), rate of change (10% and 50%), strength of change (0.5 SD and 2 SD), direction of change (polarization and secular change) and reliability of change (60% and 90%). For each of these conditions, we generated 100 simulated data (ending up with 3,200 datasets). We then applied our procedure of classification both using the fixed effects and the mixed effects modeling approaches, ending up with a distribution of accuracy scores per condition and modeling approach. In the final step, we tested the differences in accuracy scores using a simple t-test for each comparison.

We see that, under most conditions, a fixed effects approach fares better. This becomes particularly notable when the rate of change is low—considering that partial pooling shrinks estimates to global mean, a low rate of change under a multilevel model means that estimates become indistinguishable across cases. This implies that our conclusions stand under a different assumption.¹

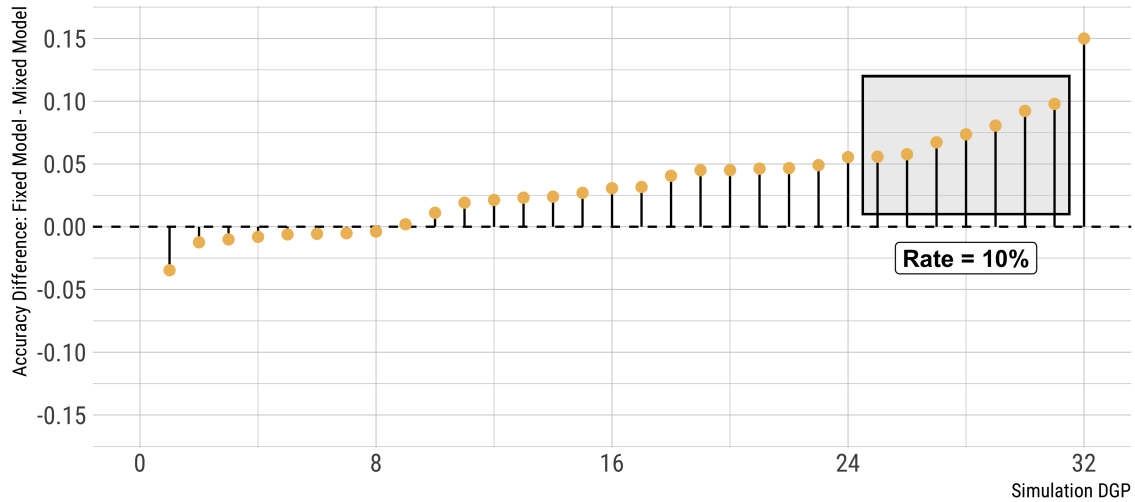


Figure D1: Comparisons Between Fixed Model Approach and Mixed Model Approach
 Notes: The figure depicts the difference scores between the fixed model approach and the mixed model approach.

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- Lersch, Philipp M. 2023. "Change in Personal Culture over the Life Course." *American Sociological Review* 88(2):252–83. doi: 10.1177/00031224231156456.

¹Of course, there may be alternative modeling approaches such as mixture models or latent transition models that may allow us to improve our estimates. We urge future studies to improve upon this point.